

```
from tensorflow.keras.datasets import mnist
```

```
(mnist_x, _), _ = mnist.load_data()
```

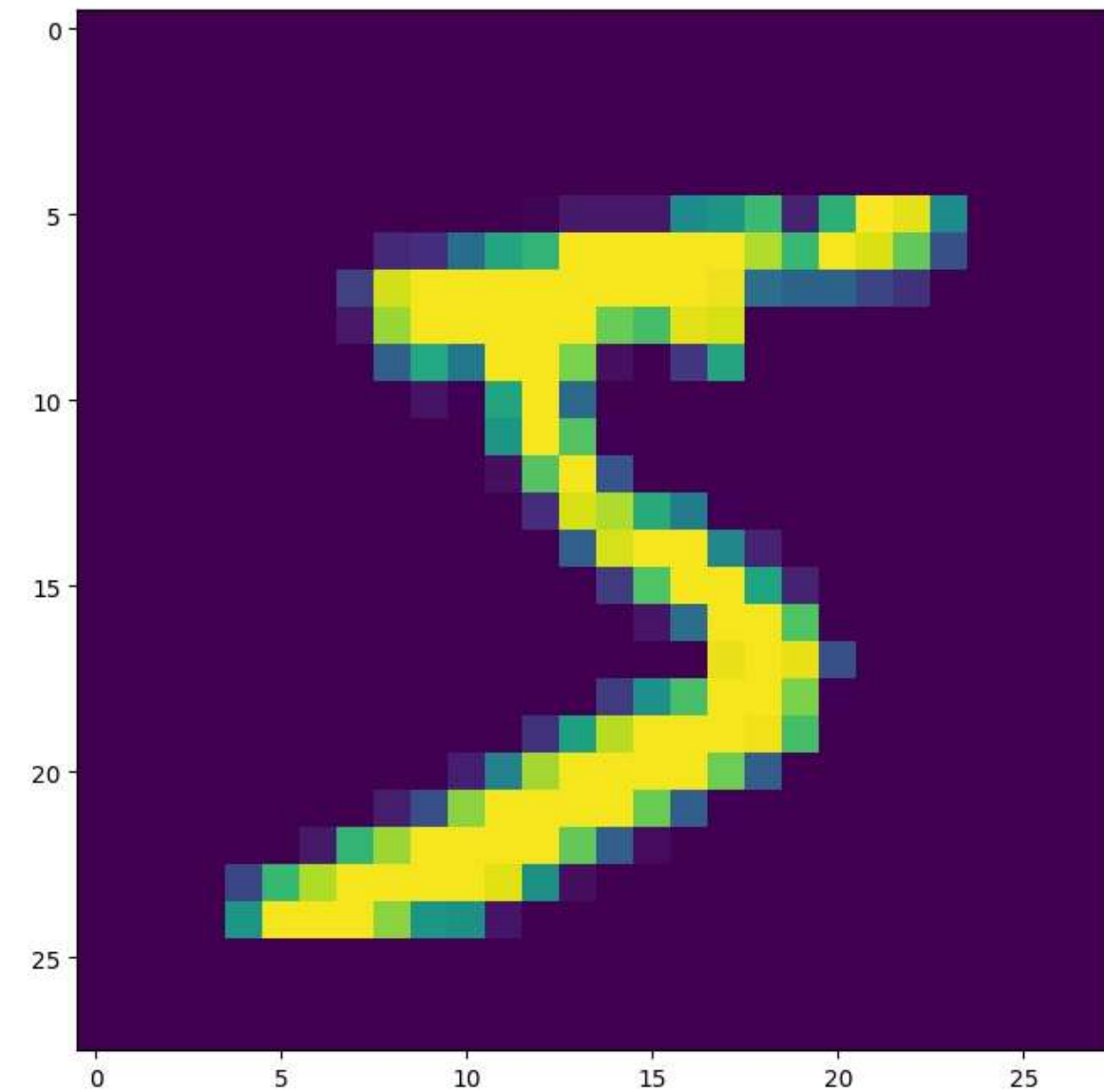
Downloading data from <https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz>
11490434/11490434 ————— 0s 0us/step

```
print(mnist_x.shape)
```

```
(60000, 28, 28)
```

```
import numpy as np
import matplotlib.pyplot as plt

plt.figure(figsize=(10, 8))
plt.imshow(mnist_x[0])
plt.show()
patch = mnist_x[0][5:9, 9:12]
patch
```



```
array([[ 0,  0,  0],
       [36, 94, 154],
       [253, 253, 253],
       [253, 253, 253]], dtype=uint8)
```

```
filter_kernal= np.array([
    [1, 1],
    [1, -1]
```

```

])
top_left_patch = patch[0:2, 0:2]

print(top_left_patch)

multiplyMatrix = top_left_patch * filter_kernel
print(multiplyMatrix)

top_left_patch_out = np.sum(multiplyMatrix)

```

```

[[ 0  0]
 [36 94]]
[[ 0  0]
 [ 36 -94]]

```

```

def convolutoin(image_matrix, kernel):
    img_height, img_width = image_matrix.shape[0], image_matrix.shape[1]

    kernel_height, kernel_width = kernel.shape[0], kernel.shape[1]

    out_height, out_width = kernel_height + 1, kernel_width + 1

    feature_map = np.zeros((out_height, out_width))

    for r in range(out_height):
        for c in range(out_width):

            curr_patch = image_matrix[r: r + kernel_height, c:c + kernel_width]

            multi = curr_patch * kernel

            total = np.sum(multi)

            feature_map[r][c] = total
    return feature_map

```

```

feature_map = convolutoin(patch, filter_kernel)

plt.imshow(feature_map)
plt.axis('off')

```

```
plt.show()
```

